

1 AdS/CFT Correspondence

A first concrete example of the holographic principle was given by Juan Maldacena in 1997 [**Maldacena1997**], when he conjectured the equivalence between a $\mathcal{N} = 4$ supersymmetric Yang-Mills theory with conformal symmetry and string theory on $AdS_5 \times S^5$, the so-called AdS/CFT correspondence. This was motivated by the fact, that by taking specific limits of type IIB string theory with D3-branes one can get either one of the two theories (see [**DHoker2002**] for a detailed review). But one does not have to study string theory to motivate the correspondence. The equivalence will already become apparent by looking at their respective symmetry groups. Since the two theories are conjectured to be the same, their symmetry group have to coincide. In general, the conformal field theory lives in a d dimensional flat spacetime and the gravity theory is defined in a $d + 1$ dimensional hyperbolic background. A hyperbolic space is usually defined as a hypersurface of a higher dimensional flat space. This means that an AdS_5 space inherits the Lorentz $SO(4, 2)$ symmetry from its embedding, which has 15 generators. The symmetries of a CFT on the other hand are generated by the usual 10 Lorentz generators corresponding to translations, rotations and boosts with additional 5 conformal symmetries, which include four special conformal and one scale transformation. These 15 generators satisfy the same algebra as the isometry group of the AdS space. In fact, $SO(4, 2)$ is also called the conformal group. On the gravity side we further have a $SO(6)$ symmetry from the compactification on the sphere S^5 . This corresponds to $\mathcal{N} = 4$ supersymmetry of the field theory, since $SO(6) \cong SU(4)$.

The realization of holography lies in the connection of the background spacetime on which the two theories are defined. Take the metric in a coordinate patch [**Bayona2005**], which covers half of AdS_{d+1} and can be obtained by coordinate transformation and conformal rescaling of the global coordinates

$$ds^2 = \frac{L^2}{\cos^2 \theta} (-d\tau^2 + d\theta^2 + \sin^2 \theta d\Omega^2) \quad (1.1)$$

with $0 \leq \tau < 2\pi$, $0 \leq \theta < \frac{\pi}{2}$ and $\sum_{i=1}^d \Omega_i^2 = 1$. We see that the boundary lies at $\theta = \phi/2$ and has the topology of $\mathbb{R} \times S^{d-1}$. We can also get this space from a conformal

compactification of Minkowski space, on which the field theory is defined. Therefore it is sensible to say, that the CFT lives on the conformal boundary of AdS_{d+1} or that the entire information of the gravity theory can be encoded in its boundary.

1.1 Boundary values as sources

A more precise formulation of the correspondence was given by Witten [Witten1998]. Assume that for operators in the CFT there exist a term in the action that couples them to an external source J .

$$\int dx \mathcal{O}(x) J(x) \quad (1.2)$$

The usual procedure for a QFT is to construct a partition function $Z[J]$ and then calculate the n -point functions as functional derivatives of $Z[J]$ with respect to the external field J . In the following we will express the bulk action as a functional of the boundary values and calculate the two-point function for the field theory. For simpler calculations and also implementation consider an AdS space with Euclidean metric, the hyperbolic space $\mathbb{H}^{d+1}$, as the bulk with metric

$$ds^2 = \frac{L^2}{z^2} (dz^2 + \eta_{ij} dx^i dx^j) \quad (1.3)$$

where the boundary lies at $z = 0$. In this work the boundary coordinates are usually denoted by $x, y \in \mathbb{R}^d$ and the holographic direction by $z \in \mathbb{R}$. Consider the case of a free complex scalar field in this curved background

$$S_{\text{bulk}}[g_{\mu\nu}, \phi] = \int d^d x dz \sqrt{g} [g^{\mu\nu} \partial_\mu \phi^* \partial_\nu \phi + m^2 |\phi|^2 - R] \quad (1.4)$$

After integration by parts for ϕ -dependent terms we get

$$S_{\text{bulk}}[\phi] = - \int d^d x dz \sqrt{g} \phi^* (-\square_g + m^2) \phi + \int d^d x dz \partial_\mu (\sqrt{g} g^{\mu\nu} \phi^* \partial_\nu \phi) \quad (1.5)$$

with the Laplacian

$$\begin{aligned} \square_g &= \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} g^{\mu\nu} \partial_\nu) \\ &= \frac{z^2}{L^2} (\partial_z^2 + (1-d) \frac{1}{z} \partial_z + \partial_i^2) \end{aligned} \quad (1.6)$$

The second term can be expressed as a surface integral over the boundary with induced metric h_{ij} and normal vector n^μ

$$\int d^d x \sqrt{h} \phi^* n^\mu \partial_\mu \phi = \int d^d x \frac{1}{z^{d-1}} \phi^* \partial_z \phi \quad (1.7)$$

To evaluate the derivative $\partial_z \phi$ we look for a classical solution for ϕ . By imposing $\delta S = 0$ we find the equation of motion

$$z^2 \partial_z^2 \phi - (d-1)z \partial_z \phi + (z^2 \partial_x^2 - L^2 m^2) \phi = 0 \quad (1.8)$$

The solution will be a convolution of the Green's function or propagator K satisfying

$$(-\square_g + m^2)K(z) = \delta(z) \quad (1.9)$$

To find an expression for K we look for a solution of homogeneous equation, which has a singularity at one point on the boundary. This corresponds to the behaviour we expect for a δ -function. Since we have translation invariance in x direction the Green's function should only be a function of z . Consider

$$K(z) = z^\alpha \quad (1.10)$$

and use inversion symmetry to map $x \rightarrow \frac{x}{z^2 + |x|^2}$ such that $K(z)$ becomes

$$K(z) = \frac{z^\alpha}{(z^2 + |x|^2)^\alpha} \quad (1.11)$$

This function has a singularity at $x = 0$ for $z \rightarrow 0$ which means it behaves like a δ function on the boundary. We can now express our solution as a convolution of the Green's function with a source

$$\phi(x) \sim \int d^d x K(z, x-y) \phi_0(y) \quad (1.12)$$

where $\phi_0(x) \equiv \phi(x, z=0)$. The derivative with respect to z is then to lowest order in z

$$\partial_z \phi \sim z^{d-1} \int d^d y \frac{\phi_0(y)}{|x-y|^{2\alpha}} \quad (1.13)$$

With this the on-shell action will be of the form

$$S_{\text{bulk}}^{\text{cl}}[\phi_0] = \int d^d x d^d y \frac{\phi_0^*(x) \phi_0(y)}{|x-y|^{2\alpha}} \quad (1.14)$$

A possible coupling of a complex boundary field ψ with source ϕ_0 is

$$- \int \phi_0^* \psi + \psi^* \phi_0 \quad (1.15)$$

Since the AdS/CFT correspondence states that the boundary and bulk theories are the

equivalent, their partition function should contain the same information, i.e.

$$Z_{\text{QFT}} = Z_{\text{bulk}} = e^{-S_{\text{bulk}}^{\text{cl}}[\phi_0]} \quad (1.16)$$

The classical bulk action becomes the generating functional for the field theory and we can compute the two-point function for ψ by taking a functional derivatives

$$\begin{aligned} \langle \psi^*(x) \psi(y) \rangle &= \frac{\delta}{\delta \phi_0(x)} \frac{\delta}{\delta \phi_0^*(y)} Z[\phi_0] \\ &\sim \frac{1}{|x - y|^{2\alpha}} \end{aligned} \quad (1.17)$$

Thus we have found a polynomial decay of the correlation function. In this fashion one can calculate any n-point function of the QFT without any precise knowledge of its action or dynamics, since the information is provided by the bulk theory. One should note, that this is the weakest form of the AdS/CFT correspondence, because it is only valid in the saddle point approximation of the bulk theory, i.e. the supergravity limit of string theory. However the equivalence is believed to hold even in the quantum regime and on all energy scales.

1.2 Conformal two-point function

Due to the large group of conformal transformation the general form of the one, two and three-point functions are completely determined by symmetry considerations up to constant factors. By calculating the two-point correlation for two complex scalar field ψ_1, ψ_2 with conformal dimensions Δ_1, Δ_2 we show that the previous results in fact produced the correct behaviour.

In general the two-point function will be a function of the distance between two fields because of translational invariance

$$\langle \psi_1^*(x) \psi_2(y) \rangle = f(|x - y|) \quad (1.18)$$

The fields behave under a scaling transformation $x \rightarrow \lambda x$ like

$$\psi_i(\lambda x) = \lambda^{\Delta_i} \psi_i(x) \quad (1.19)$$

Since we have an invariance under such rescaling we can further restrict the form

$$\begin{aligned} \langle \lambda^{\Delta_1} \psi_1^*(\lambda x) \lambda^{\Delta_2} \psi_2(\lambda y) \rangle &= \lambda^{\Delta_1 + \Delta_2} f(\lambda(x - y)) \\ &\stackrel{!}{=} f(x - y) \end{aligned} \quad (1.20)$$

Then f has to decay according to a power-law

$$f(|x - y|) \sim \frac{1}{|x - y|^{\Delta_1 + \Delta_2}} \quad (1.21)$$

Another symmetry to consider is the invariance under inversion

$$x \rightarrow \frac{x}{|x|^2} \quad (1.22)$$

This will give us a relation of the weights. We impose

$$\begin{aligned} \left\langle \frac{1}{x^{2\Delta_1} y^{2\Delta_2}} \psi_1^* \left(-\frac{1}{x}\right) \psi_2 \left(-\frac{1}{y}\right) \right\rangle &= \frac{1}{x^{2\Delta_1} y^{2\Delta_2}} \frac{c}{\left(-\frac{1}{x} + \frac{1}{y}\right)^{\Delta_1 + \Delta_2}} \\ &\stackrel{!}{=} \frac{c}{|x - y|^{\Delta_1 + \Delta_2}} \end{aligned} \quad (1.23)$$

To fulfill this condition we have to set $\Delta_1 = \Delta_2$ and arrive at the correlation between two scalar fields

$$\langle \psi^*(x) \psi(y) \rangle \sim \frac{1}{|x - y|^{2\Delta}} \quad (1.24)$$

This is in agreement with the result from the correspondence. By comparing the two expressions the exponent α can be identified with the conformal dimension of the boundary field Δ . This gives us a practical way to check if a theory is not a CFT. If the field theory in question is not in a conformal invariant phase, the two-point function will usually decay exponentially with distance.

1.3 Holographic renormalization

When calculating observables of a QFT one in general has to do them carefully in a renormalized fashion, as quantities can easily diverge due to the presence of arbitrary small scales or infinite volumes in the definition of the bare action. A potential divergence is for example the integration $\int d^d x$ over the boundary at $z = 0$ as the area is infinite. So before we perform any calculation, we have to regularize the theory by introducing a cutoff to the length or energy scale, e.g. integral goes over a surface at some $z_0 \neq 0$. The cutoff will then be sent $z \rightarrow 0$ after the calculation. This transformation is part of the isometry $(x, z) \rightarrow (\lambda x, \lambda z)$ of AdS space, which is dual to the dilation symmetry of CFTs. For the boundary theory this corresponds to a coarse-graining, that will induce a RG flow.

As an example let's consider a massive field in the bulk. A typical two-point function between two points in the bulk at $z = z_0$ will decay exponentially with distance

[Fukuma2002]

$$\langle \phi^*(x)\phi(y) \rangle_{z=z_0} \sim \sum_{\text{paths}(x,y)} e^{-m\mathcal{D}(x,y)} \quad (1.25)$$

This expression will be dominated by the geodesic connecting the points (x, z_0) and (y, z_0) for $m \gg L^{-1}$, so

$$\langle \phi^*(x)\phi(y) \rangle_{z=z_0} \sim e^{-m\mathcal{D}_{\text{geo}}(x,y)} \quad (1.26)$$

The geodesic in AdS will be of logarithmic dependence on $|x - y|$

$$\mathcal{D}_{\text{geo}}(x, y) = L \ln \left(\frac{\left(|x - y| + \sqrt{|x - y|^2 + z_0^2} \right)^2}{z_0^2} \right) \quad (1.27)$$

We see now that the two-point function will be a power-law at distances bigger than z_0

$$\langle \phi^*(x)\phi(y) \rangle_{z=z_0} \sim \frac{1}{|x - y|^{2mL}} \quad \text{for } |x - y| \gg z_0 \quad (1.28)$$

which is the familiar form we get for a CFT and we can identify $\Delta = mL$. However, the behaviour at short scale is modified and is not polynomial anymore. This means that the long distance physics of the bulk theory is related to the UV of the boundary. This is called the IR-UV relation, which was first discussed in [Susskind1998]. The theory that we find at z_0 is renormalized at energy scale

$$\Lambda_0 = \frac{1}{z_0} \quad (1.29)$$

This implies a relation of the renormalization group flow with the radial flow direction of the bulk. In a Wilsonian approach we would integrate out the UV degrees of freedom in the CFT, which should correspond to an integration of the long wavelength modes in the bulk [Heemskerk2010]. In classical limit we can then define an inner correspondence of the CFT at z_0 and the interior volume of AdS space by integrating out fields in the excluded boundary area [Balasubramanian1998]. A more precise mapping can be achieved when one goes into the Hamilton-Jacobi formulation of the gravity theory, where the first order flow equations can be cast into the form of Callan-Symanzik equation [Boer1999]. In a semiclassical treatment the field values $\phi(x, z)$ can be interpreted as the running coupling constants of the CFT for a perturbation $\phi(x, \epsilon) = \epsilon^{d-\Delta} \tilde{\phi}(x)$. Hamilton's equations are then equivalent to the renormalization group flow equations [Balasubramanian2012]

$$z \frac{d}{dz} \phi = \frac{\delta \mathcal{H}}{\delta \pi} \leftrightarrow \Lambda \frac{d}{d\Lambda} g = \beta(g) \quad (1.30)$$

where $\mathcal{H}$ is the Hamiltonian with conjugate momentum π and g is some running coupling constant in the CFT. The metric of the bulk is therefore a metric of the space of RG flow. However, such a mapping is only valid in the semiclassical limit and also depends on the renormalization scheme. That means that one has to take precautions when setting up a coarse-graining procedure for a CFT, since not every renormalization will exhibit a geometric structure even if there exist a dual gravity theory for the CFT.